

Write $\varepsilon^j = R^j - u_h^j$, and let $\tilde{R}$ and u_h be the continuous, piecewise affine interpolants of their nodal values. Set $\varepsilon(t) = \tilde{R}(t) - u_h(t)$. The elliptic-reconstruction bounds give

$$\|\varepsilon^0\|_{\infty, \Omega} \leq \eta_{\text{ell}}^0, \quad \|\delta_t \varepsilon^j\|_{\infty, \Omega} \leq \eta_{\text{ell}, \delta_t}^j.$$

For $j \geq 1$, the discrete equation and the definition of ψ_h^j imply $\psi_h^j = -\delta_t u_h^j$. Moreover, because $\mathcal{M}$ is time-independent and linear, the defining elliptic problems yield, on I_j ,

$$\mathcal{M}\tilde{R}(s) = f^j + \psi_h^j + (s - t_j)\delta_t(f^j + \psi_h^j), \quad \partial_s \tilde{R}(s) = \delta_t R^j.$$

Consequently, for $w = u - \tilde{R}$,

$$\mathcal{K}w(s) = f(s) - f^j + (t_j - s)\delta_t(f^j + \psi_h^j) - \delta_t \varepsilon^j, \quad s \in I_j.$$

The function w belongs to the space required by the Green's-function representation lemma, and $w(0) = u_0 - u_h^0 - \varepsilon^0$. Applying that lemma at T , taking absolute values, and using the Green's-function L^1 bound therefore gives

$$\begin{aligned} \|w(T)\|_{\infty, \Omega} &\leq \phi_{0, \Gamma}(T)(\|u_0 - u_h^0\|_{\infty, \Omega} + \eta_{\text{ell}}^0) \\ &\quad + \sum_{j=1}^M \int_{I_j} \phi_{0, \Gamma}(T - s) \|f(s) - f^j + (t_j - s)\delta_t(f^j + \psi_h^j)\|_{\infty, \Omega} ds \\ &\quad + \sum_{j=1}^M \eta_{\text{ell}, \delta_t}^j \int_{I_j} \phi_{0, \Gamma}(T - s) ds. \end{aligned}$$

Finally, $u_h(T) = u_h^M$, $\tilde{R}(T) = R^M$, and $\|R^M - u_h^M\|_{\infty, \Omega} \leq \eta_{\text{ell}}^M$. Thus the desired a posteriori upper bound is

$$\begin{aligned} \|u(T) - u_h(T)\|_{\infty, \Omega} &\leq \eta_{\text{ell}}^M + \phi_{0, \Gamma}(T)(\|u_0 - u_h^0\|_{\infty, \Omega} + \eta_{\text{ell}}^0) \\ &\quad + \sum_{j=1}^M \left[\int_{I_j} \phi_{0, \Gamma}(T - s) \|f(s) - f^j + (t_j - s)\delta_t(f^j + \psi_h^j)\|_{\infty, \Omega} ds + \eta_{\text{ell}, \delta_t}^j \int_{I_j} \phi_{0, \Gamma}(T - s) ds \right] \end{aligned}$$

Every term on the right is specified by the data, the discrete solution, and the assumed elliptic estimators. The estimate is understood whenever its displayed maximum-norm data terms are finite.